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INVITED COMMENTARY

Interpreting Power-Force-Velocity Profiles for Individualized and Specific Training

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Recent studies have brought new insights into the evaluation of power-force-velocity profiles in both ballistic push-offs (eg, jumps) and sprint movements. These are major physical components of performance in many sports, and the methods the authors developed and validated are based on data that are now rather simple to obtain in field conditions (eg, body mass, jump height, sprint times, or velocity). The promising aspect of these approaches is that they allow for more individualized and accurate evaluation, monitoring, and training practices, the success of which is highly dependent on the correct collection, generation, and interpretation of athletes' mechanical outputs. The authors therefore wanted to provide a practical vade mecum to sports practitioners interested in implementing these power-force-velocity–profiling approaches. After providing a summary of theoretical and practical definitions for the main variables, the authors first detail how vertical profiling can be used to manage ballistic push-off performance, with emphasis on the concept of optimal force–velocity profile and the associated force–velocity imbalance. Furthermore, they discuss these same concepts with regard to horizontal profiling in the management of sprinting performance. These sections are illustrated by typical examples from the authors' practice. Finally, they provide a practical and operational synthesis and outline future challenges that will help further develop these approaches.

***Keywords**:* explosive performance, jump, sprint, team sports, athletics, strength training

One of the main physical performance determinants in sports such as athletics, rugby, soccer, football, volleyball, and basketball is the ability to produce high mechanical power output during jumps and sprint accelerations.[1,2] This power output depends on the ability of athletes' neuromuscular and osteoarticular systems to generate high levels of force, apply it with effectiveness onto the environment (ie, supporting ground, ball, projectile), and produce this force at high contraction velocity. Force and velocity are therefore considered the underpinning features of mechanical power output in sport movements.[3,4] Although the assessment and long-term monitoring of these capabilities is paramount for both performance and rehabilitation processes, such an accurate evaluation has long been associated with expensive and often laboratory-based technologies. Recently, our research group has presented simple field methods to compute force, velocity, and power output in jumping[5] and sprinting[6] calculated via measurements from widely accessible and practical devices. Thanks to these methods, all the important mechanical outputs of jumping and sprinting can be derived from basic measures of body mass, lower-limb length, jump height, and distance–time or speed–time measurements only.[6–8]

Recently, we had the opportunity to discuss the implementation of these "simple methods" with many sport practitioners, and we realized that beyond the description presented in the published papers, it was necessary to detail how to interpret the measurements for an efficient use in everyday practice. Our aim here is to provide a practical vade mecum to readers wishing to use power-force-velocity profiling for more individualized diagnostic and efficient training. The key points of this commentary will be supported by illustrations of typical data collected in our research, training, or consultancy practice over the past decade.

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Definitions

The power-force-velocity–profiling approach is based on force–velocity (F–V) and power–velocity relationships characterizing the maximal mechanical capabilities of the lower limbs' neuromuscular system. The definition and the practical interpretation of the main mechanical variables of interest are presented in Table 1.

Vertical Profiling for Ballistic Push-Off Performance

The input measurements necessary to correctly determine vertical profile[5,9,10] are the athlete's body mass, lower-limb length in fully extended position, starting height, and jump height (measured under a spectrum of loading parameters). The latter can now be easily and accurately measured using simple and accessible devices.[7,8] Jump height should be measured across repeated measurements with at least 5 additional loads (evenly ranging between 0 kg and the additional load with which the athlete is able to jump about 10 cm), after which the F–V profile and all other computations can be completed. Research conclusions show that jumping performance is determined by maximal mechanical power output (*VTC-Pmax*) and the magnitude of the relative difference between the slope of the linear F–V relationship (*Sfv*) and Sfv_{opt} for a given individual (*FVimb*).[9] Thus, in practical terms, should a training program be designed to improve athletes' ballistic push-off performance (eg, jumps, single maximal push-offs, change of direction), the focus should be placed on increasing *VTC-Pmax* and/or decreasing *FVimb*. With regard to athletes displaying significant imbalance in mechanical capacities,

Table 1 Definition and Practical Interpretation of the Main Variables of Interest When Using Power-Force-Velocity Profiling in Ballistic Push-Offs (Vertical Profiling) and Sprinting (Horizontal Profiling)

Profiling variable	Definition and computation	Practical interpretation
Vertical		
VTC-F0 (N/kg)	Theoretical maximal force production of the lower limbs as extrapolated from the linear loaded jump squats' force–velocity (F–V) relationship; *y*-intercept of the linear F–V relationship.	Maximal concentric force output (per unit body mass) that the athlete's lower limbs can theoretically produce during ballistic push-off. Determined from the entire F–V spectrum, it gives more integrative information on force capability than, eg, concentric squat 1-repetition-maximum load.
VTC-V0 (m/s)	Theoretical maximal extension velocity of the lower limbs as extrapolated from the linear loaded jump squats' F–V relationship; *x*-intercept of the linear F–V relationship.	Maximal extension velocity of the athlete's lower limbs during ballistic push-off. Determined from the entire F–V spectrum and very difficult, if not impossible, to reach and measure experimentally. It also represents the capability to produce force at very high extension velocities.
VTC-Pmax (W/kg)	Maximal mechanical power output, computed as $Pmax = F0 \times V0/4$ or as the apex of the P–V 2nd-degree polynomial relationship.	Maximal power output capability of the athlete's lower-limb neuromuscular system (per unit body mass) in the concentric and ballistic extension motion.
S*fv*	Slope of the linear F–V relationship, computed as $Sfv = -F0/V0$.	Index of the athlete's individual balance between force and velocity capabilities. The steeper the slope, the more negative its value, the more "force-oriented" the F–V profile, and vice versa.
Sfv_{opt}	For a given push-off distance, body mass, and *Pmax*, the unique value of *Sfv* that maximizes jump height. For detailed computation, see Appendix in Samozino et al.[10]	The optimal F–V profile that represents the optimal balance, for a given individual, between force and velocity capabilities. For a given maximal power *Pmax,* this profile will be associated, ceteris paribus, with the highest ballistic push-off performance possible for this individual. Training programs should be designed to both increase *Pmax* and orient *Sfv* toward Sfv_{opt}.
FVimb (%)	Magnitude of the relative difference between *Sfv* and Sfv_{opt} for a given individual. Computed as $(Sfv/Sfv_{opt}) \times 100$ and expressed in percentage.	Magnitude of the difference between actual and optimal F–V profiles. A value of 100% means $Sfv = Sfv_{opt}$, ie, optimized F–V profile. Values above 100% mean an imbalance with a deficit in velocity, and vice versa. The larger the difference with the optimal 100% value, the larger the imbalance.
Horizontal		
HZT-F0 (N/kg)	Theoretical maximal horizontal force production as extrapolated from the linear sprint F–V relationship; *y*-intercept of the linear F–V relationship.	Maximal force output (per unit body mass) in the horizontal direction. Corresponds to the initial push of the athlete onto the ground during sprint acceleration. The higher the value, the higher the sprint-specific horizontal force production.
HZT-V0 (m/s)	Theoretical maximal running velocity as extrapolated from the linear sprint F–V relationship; *x*-intercept of the linear F–V relationship.	Sprint-running maximal velocity capability of the athlete. Slightly higher than the actual maximal velocity. The theoretical maximal running velocity the athlete would be able to reach should mechanical resistances (ie, internal and external) against movement be null. It also represents the capability to produce horizontal force at very high running velocities.
HZT-Pmax (W/kg)	Maximal mechanical power output in the horizontal direction, computed as $Pmax = F0 \times V0/4$, or as the apex of the P–V 2nd-degree polynomial relationship.	Maximal power-output capability of the athlete in the horizontal direction (per unit body mass) during sprint acceleration.
RF (%)	Ratio of force, computed as the ratio of the step-averaged horizontal component of the ground-reaction force to the corresponding resultant force.	Direct measurement of the proportion of the total force production that is directed in the forward direction of motion, ie, the mechanical effectiveness of force application of the athlete. The higher the value, the more important the part of the total force output directed forward.
RFmax (%)	Maximal value of *RF*, computed as maximal value of *RF* for sprint times >0.3 s.	Theoretically maximal effectiveness of force application. Direct measurement of the proportion of the total force production that is directed in the forward direction of motion at sprint start.
D_{RF}	Rate of decrease in *RF* with increasing speed during sprint acceleration, computed as the slope of the linear *RF*–V relationship.	Describes the athlete's capability to limit the inevitable decrease in mechanical effectiveness with increasing speed, ie, an index of the ability to maintain a net horizontal force production despite increasing running velocity. The more negative the slope, the faster the loss of effectiveness of force application during acceleration, and vice versa.

training programs should prioritize training the lacking mechanical capability to shift *Sfv* toward Sfv_{opt}. The main interest of the current approach is that the diagnostics, and resultant training periodization, are individualized and easily monitored. Consequently, the ability to frequently monitor these outputs permits the analysis of changes in *VTC-Pmax* and *FVimb* over time (eg, once every month) and can assist in the targeted implementation and reimplementation of efficient and dynamic programming practices. The first case report (Figure 1) illustrates this with data from 2 athletes with a similar push-off distance. Although athlete A has a higher *VTC-Pmax,* his squat-jump performance is lower because he has an F–V imbalance. Athlete B has a lower *VTC-Pmax,* but his profile is almost exactly equal to his individual optimal profile (only 1% imbalance). The current approach would suggest, therefore, that athlete A's training should prioritize the development of maximal force capabilities to correct his imbalance and increase *VTC-Pmax*. Once this goal is achieved, he may transition into training similar to that of athlete B, to improve his *VTC-Pmax* while maintaining his corrected (ie, optimal) profile.

The second example shows 2 young players from the same soccer team (French first-league professional club academy U19). As shown in Figure 2, these players have quite similar *VTC-Pmax* and Sfv_{opt} values but display opposing *FVimb* characteristics: Player A shows a force deficit, whereas player B shows a velocity deficit. Furthermore, the absolute difference with their respective Sfv_{opt} is lower in player B than in player A (28% vs 37%). This relatively smaller *FVimb* and slightly higher *VTC-Pmax* in player B explain his higher squat-jump performance.

With this result in mind, this approach suggests that the most efficient way to train and improve ballistic push-off performance in both these players would be an individualized program (indexed on each player's *FVimb*) that targets the development of different

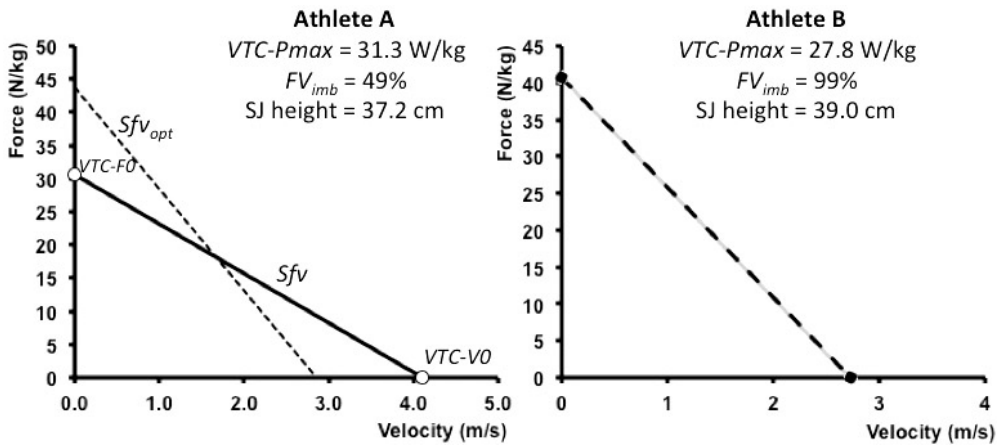

Figure 1 — Vertical force–velocity profiles of 2 track and field athletes (body mass for A, 67.2 kg, and B, 82.8 kg; push-off distance for A, 0.34 m, and B, 0.35 m) obtained from maximal squat jumps (SJ) against additional loads of 0, 10, 20, 30, and 40 kg. Despite a higher *VTC-Pmax* (maximal mechanical power output) value, athlete A's squat-jump performance is lower because his FV_{imb} (magnitude of the relative difference between the slope of the linear force–velocity relationship [*Sfv*] and Sfv_{opt}) is greater than for athlete B. For athlete A, the black line indicates the actual profile, and the dashed line, the optimal profile. Note that athlete B's profile if almost optimal, and therefore the actual and optimal relationships are confounded in the right panel (gray line and black dashed line). Abbreviations: *VTC-F0,* maximal force production of the lower limbs; *VTC-V0,* maximal extension velocity of the lower limbs.

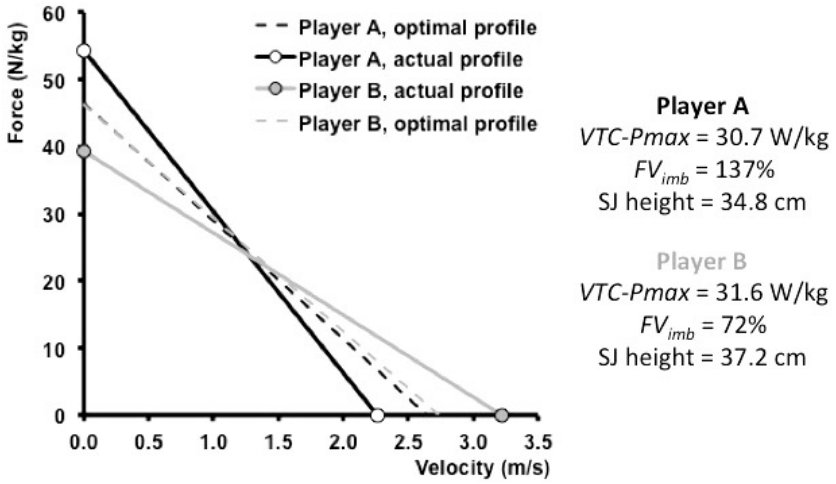

Figure 2 — Vertical force–velocity profiles of 2 elite young (under-19) soccer players (body mass for A, 78 kg, and B, 75.5 kg; push-off distance for A, 0.26 m, and B, 0.28 m) obtained from maximal squat jumps (SJ) against additional loads of 0, 10, 20, 40, and 50 kg. One player has a force deficit (magnitude of the relative difference between the slope of the linear force–velocity relationship [*Sfv*] and Sfv_{opt} [FV_{imb}] of 72%), whereas the other has a velocity deficit (FV_{imb} of 137%). Player A is a central defender and player B is a goalkeeper. Abbreviation: *VTC-Pmax,* maximal mechanical power output.

capabilities. Our yet-unpublished observations have shown that such an individually optimized approach is more efficient that a one-size-fits-all program, identical for those 2 players.

The latter example raises an important question, however, with regard to the application of improving ballistic push-off performance in cyclic movements such as sprint running. This particular question is the main interest when developing forward (sprint) acceleration and performance characteristics, for instance, in soccer or rugby players (except for some players like goalkeepers or some specific sport actions involving jumps), and will be discussed in the following section detailing horizontal profiling for sprint performance.

Horizontal Profiling for Sprint Performance

The inputs that must be measured to determine the horizontal profile[6] are the athlete's body mass and height and either distance–time or speed–time running data. The latter can be measured using a series of timing gates (at least 5 split times, eg, 5, 10, 20, 30, and 40 m) or a laser or radar device (eg, ~50-Hz Stalker ATSII radar, Applied Concepts Inc, Plano, TX). Wind speed, ambient temperature, and pressure must also be known to accurately estimate air-friction force. The entire power-force-velocity profile can then be computed from the simple modeling of the derivation of the speed–time curve that leads to horizontal acceleration data. Likewise, the mechanical effectiveness of force application can be determined via the linear relationship between ratio of force (*RF*) and running velocity[11] (Figure 3). Our research has shown that, in addition to maximal mechanical power output in the horizontal direction (*HZT-Pmax*), 100-m performance was related to the ability to apply high amounts of force in the horizontal direction (*RF* and rate of decrease in *RF* [D_{RF}] indices).[11–13] With regard to shorter sprints (ie, acceleration-only phases, eg, up to 10–20 m in rugby or soccer specialists), recent results have shown that the shorter the distance considered, the higher the relationship between sprint performance and maximal horizontal force production (*HZT-F0*) (unpublished observations). Thus, in practical terms, if a training program is designed to improve sprint-acceleration performance, the focus should be placed on increasing *HZT-Pmax* by improving its components (*HZT-F0* and maximal running velocity [*HZT-V0*]). This could be done by first comparing the relative strengths and weaknesses in each player's profile with the rest of the team, and then programming the training content depending on the distance over which sprint acceleration should be optimized. As for vertical profiling, the main value of this approach is that the diagnostic and subsequent targeted training interventions are individualized, and frequent monitoring of program-induced changes in *HZT-Pmax* and its mechanical determinants can make this program more efficient and dynamic in terms of adaptation to individual changes over time. In particular, since *HZT-F0* and *RF* are paramount for short sprint-acceleration performances, coupling the vertical profiling to the horizontal profiling can help identify the determinants of *HZT-F0*. Using this approach, we consider *HZT-F0* to result from the interaction of the overall strength capability of the athlete at each lower-limb extension (as assessed by the vertical profile) and his or her ability to transfer this overall strength level to the specific, forward sprint motion at the first steps (as evidenced by *RFmax*) or at steps at high velocities (as evidenced by D_{RF}) (Table 1, Figure 4). In short, a high *HZT-F0* can result from high *VTC-Pmax* and a high quality of vertical-to-horizontal transfer (ie, good *RFmax* and D_{RF} values), whereas a low *HZT-F0* can result from a high *VTC-Pmax* with a low-quality transfer (poor *RFmax* and D_{RF} values); vice versa, a low *VTC-Pmax* with a high-quality transfer (good *RFmax* and D_{RF} values); or any possible intermediate combination.

The case report used to illustrate these points shows data from 2 players of an elite rugby union team. Figure 3 shows that the 2 players have similar 20-m times (maximal acceleration from a standing start) and *HZT-Pmax* values, yet with opposite F–V profiles and *RF*-velocity profiles. Indeed, player C has higher horizontal force-production capabilities (in the specific context of sprint push-off), especially at the beginning of the sprint and notably due to a higher effectiveness of ground-force application (indicated in a higher *RFmax*). However, his D_{RF} is more negative, meaning his higher initial effectiveness decreases at a greater rate as speed increases than for player D. This has likely contributed to higher velocity

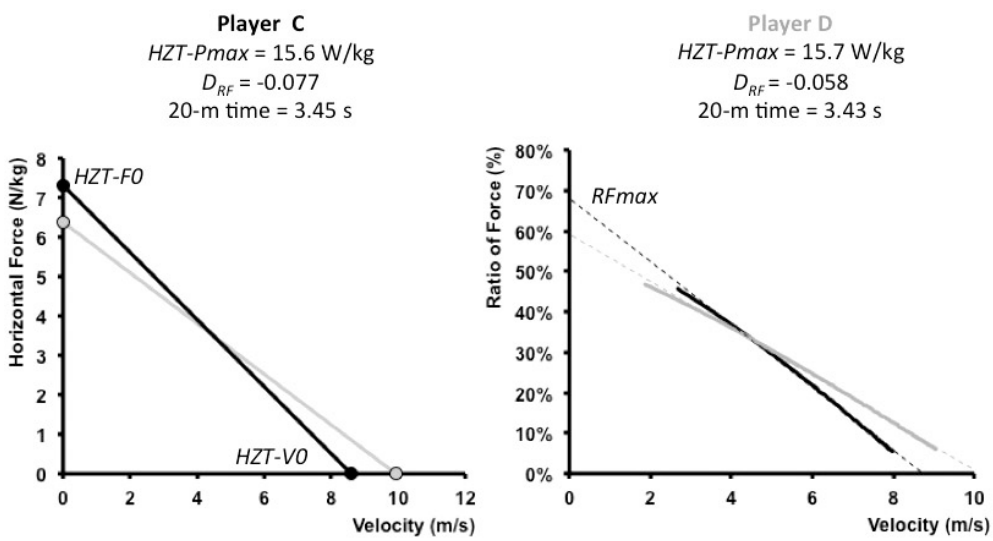

Figure 3 — Horizontal force–velocity profiles of 2 elite rugby union players (body mass for C, 108.8 kg, and D, 86.1 kg) obtained from maximal 30-m sprints. Both players reached their maximal running speed before the 30-m mark. Abbreviations: *HZT-Pmax,* maximal mechanical power output in the horizontal direction; D_{RF}, rate of decrease in ratio of force with increasing speed during sprint acceleration; *HZT-F0,* maximal horizontal force production; *HZT-V0,* maximal running velocity.

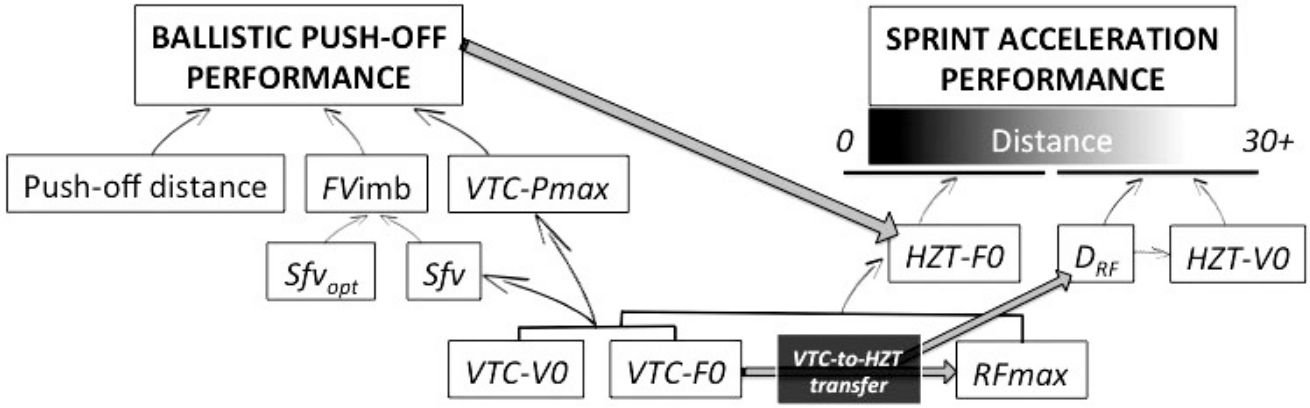

Figure 4 — Decision tree to interpret the force-velocity-power profiles in relationship with ballistic push-off (eg, jumping) and sprinting performances. These mechanistic relationships are based on both the theoretical features of our models[5,6,9–13,20] and some experimental evidence ([18,19] and unpublished data). In sprinting, the shorter the acceleration distance, the higher the importance of *HZT-F0* capabilities compared with *HZT-V0,* and vice versa. Abbreviations: FV_{imb}, magnitude of the relative difference between the slope of the linear force–velocity relationship (*Sfv*) and Sfv_{opt}; *VTC-Pmax,* maximal mechanical power output; *HZT-F0,* maximal horizontal force production; D_{RF}, rate of decrease in the ratio of force with increasing speed during sprint acceleration; *HZT-V0,* maximal running velocity; *RFmax,* maximal ratio of force.

capabilities, which explains the higher *HZT-V0* of player D. As for ballistic push-off, we suggest that the training program designed to improve sprint performance (eg, here 20-m time) in each of these 2 players should target different capabilities. A similar program given to these players (which is current practice in the majority of teams, based on our perception) will very likely result in suboptimal adaptations for both of them. In particular, player D's training should target as a priority his *HZT-F0* capabilities. Here, in terms of injury prevention, this suggests that this player could be given more strength and horizontal strength work than others and probably less maximal sprint velocity work. This could directly reduce the risk for sprinting-related injuries for this player by reducing the total time he would be exposed to high-speed running.[14–17] For this player, compared with player C (and potentially compared with the average value of the group/team), *HZT-F0* should be developed, especially through increasing *RFmax*. Adding the previously described vertical profiling to this horizontal profiling could help better determine whether a lower *HZT-F0* is due to an overall deficit of lower-limb strength (as indicated by a low *VTC-Pmax*) and/or a deficit in the transfer of this strength in the specific horizontal push-off motion (technical capability). Differences in horizontal profiles have been reported in elite rugby players according to individual player positions[18] and in young soccer players.[19]

Practical Synthesis

Figure 4 shows a decision tree, with a specific focus on ballistic push-off and sprint-acceleration performance, which are 2 major physical determinants in many sports. This figure is designed to help practitioners use the vertical and horizontal profiling approach to better detect the strengths and weaknesses in their athletes and design more-effective training interventions. Vertical profiling will provide information as to what physical capabilities should be developed to improve ballistic push-off performance and as to the maximal levels of force and velocity of the athlete's neuromuscular system. Horizontal profiling will provide information as to the specific sprint-acceleration motion and as to what underlying physical or technical feature(s) mainly limit each individual's sprint performance.

Conclusion and Perspectives

These novel approaches of vertical and horizontal force-velocity-power profiling have the potential to provide sport practitioners simple, cheap, yet accurate methods for more individualized monitoring and training of physical and technical capabilities. These methods can be easily implemented on a regular basis, since they are based on common and sport-specific movements (ballistic push-offs and sprint accelerations), and can therefore be used for long-term monitoring and training processes. Furthermore, they may also be implemented in injury-prevention and -rehabilitation processes since diagnostic information will assist in better-designed sprint-related training programs, and clear differences have been observed between injured and noninjured players.[20,21]

The limitations of these approaches have been extensively discussed,[5,6,22] and the main perspective stems from the fact that these profiling methods give information as to *what* specific muscle outputs should be developed, not *how* this should be done. This will be the next challenge that we are pleased to undertake: testing and investigating the most-efficient practical (training) methods to improve each mechanical determinant of performance and further extending the current knowledge on this topic[4] using the novel approaches presented here.

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